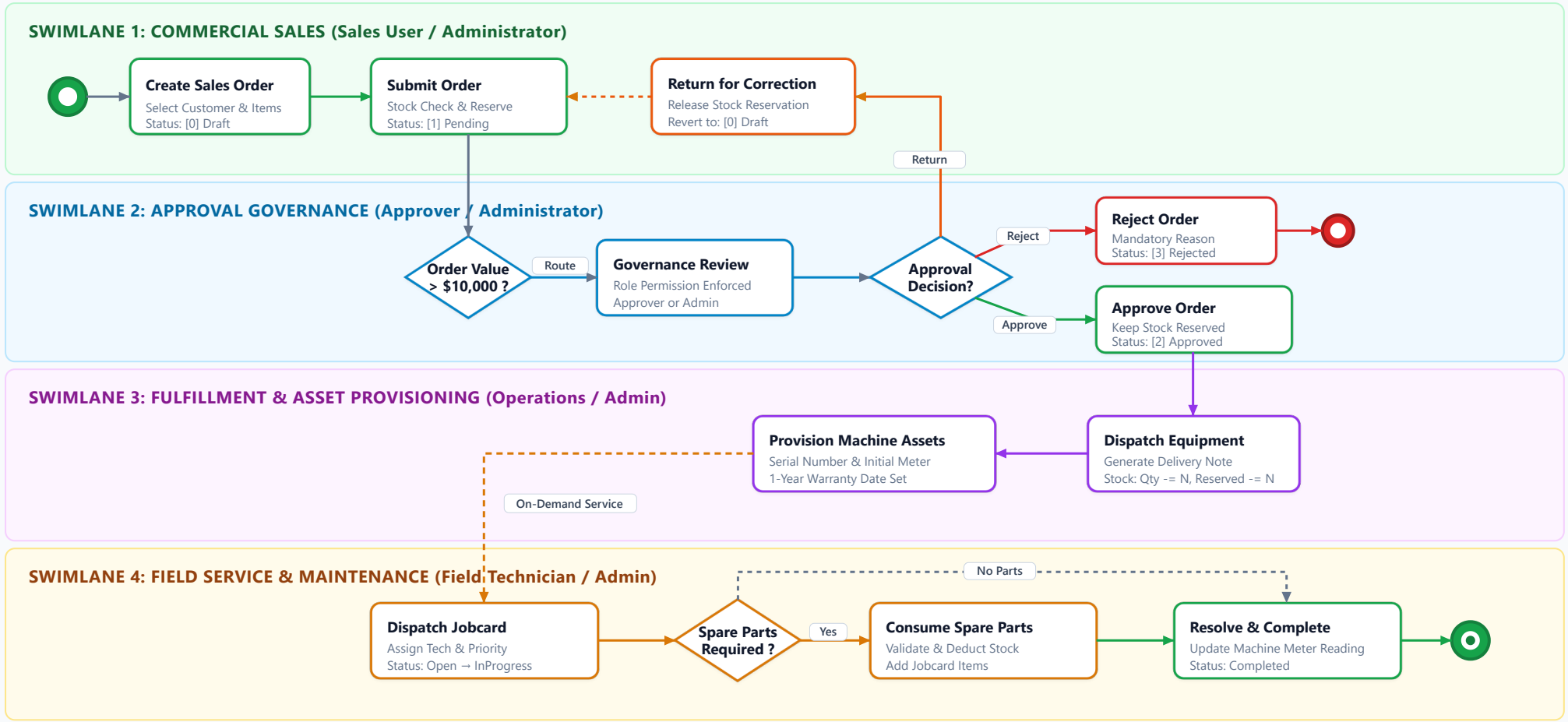
End-to-End Enterprise Operations & Activity Flow

Phase 5 Architecture: Commercial Order Governance, Warehouse Dispatch, Asset Registration & Field Service



Enterprise Governance Rules & System Architecture

Role-Based Access Control, Inventory State Guarantees, Financial Policies & Service Specifications

Role-Based Access Control (RBAC) Matrix

4 SYSTEM ROLES

Role	Allowed Actions	Scoped Data Visibility	Policy / Guard
Sales	Create draft orders, Submit for approval, Resubmit returned orders	Own sales orders only; active customers & products	SALESONLY
Approver	Approve orders (≤ $10k), Reject orders (with reason), Return for correction	All pending approval queues; order line items & history	APPROVERONLY
Administrator	All actions + High-value approvals (> $10k), Equipment dispatch, Asset provisioning	Unrestricted company-wide visibility & system audit log	ADMINONLY
Technician	Accept jobcard, Request spare parts (deduct stock), Record meter & complete	Assigned field jobcards, customer machine assets & spare parts catalog	TECHPOLICY

Financial Governance & Approval Thresholds

ENFORCED AT API & DOMAIN

1. **High-Value Authorization Ceiling ($10,000 Rule):**
   Standard `Approver` role can approve orders up to $10,000. Orders exceeding $10,000 require `Administrator` privileges; non-admin attempts are rejected with `403 Forbidden`.
2. **Mandatory Rejection Documentation:**
   Rejecting an order strictly requires a non-blank audit reason. Empty rejection reasons fail client validation and are rejected with `400 Bad Request`.
3. **Return for Correction Loop:**
   Approvers can return an order to `Draft` status with notes. All reserved warehouse stock is immediately released, allowing the sales representative to adjust quantities or pricing without deadlocks.

Physical Inventory & Reservation State Guarantees

DOUBLE-ENTRY LEDGER INTEGRITY

Lifecycle Stage	Physical Stock	Reserved Stock	Available Stock	Invariants
Order Drafted	Unchanged	Unchanged	`Stock - Reserved`	No allocation locks placed
Order Submitted	Unchanged	`Reserved += Qty`	`Stock - Reserved`	Throws `Conflict (409)` if `Qty > Available`
Order Returned / Rejected	Unchanged	`Reserved -= Qty`	Restored	Clean reservation release, no orphaned locks
Delivery Dispatched	`Stock -= Qty`	`Reserved -= Qty`	Unchanged	Generates unique Serial Number & 1-Yr Warranty
Service Spare Consumed	`Stock -= Qty`	Unchanged	`Stock - Reserved`	Instant parts allocation onto technician jobcard

Post-Sale Machinery & Field Service Flow

EQUIPMENT LIFECYCLE

- **Automated Machine Provisioning:**
  Dispatching an approved sales order generates a permanent `MachineAsset` with initial meter count, warranty expiration date (+1 year), and registered customer site.
- **Service Dispatch & Jobcards:**
  Maintenance tasks (Preventive, Breakdown, Meter Audit) dispatch a `Jobcard` to an assigned field technician with real-time status tracking (`Open → InProgress → Completed`).
- **Spare Parts Requisition & Meter Telemetry:**
  Technicians request spare parts directly into the jobcard (updating inventory in real time) and record latest machine odometer readings upon job closure.